

Robotics

Inventor Hub

Pybricks

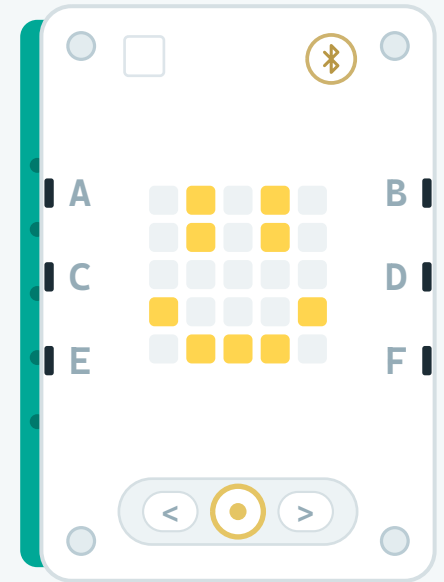
Lab 01

Autonomous Robotics & Sensor Integration

Lab 01: Line Following, Sensors & Pybricks State Machines

TA Robotics Team | Robotics & Autonomous Systems Lab

Fall 2026



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01

Hardware Setup & Hub Architecture



I Inventor Hub & Port Configuration

The Robot Inventor Hub orchestrates all sensors and actuators via high-speed UART ports:

- ◆ **DriveBase Actuators:** Dual medium angular motors configured with counter-rotation.
- ◆ **Active Perception:** Color sensor oriented down for line detection; ultrasonic sensor forward.
- ◆ **Real-time MicroPython:** Running baremetal Pybricks firmware for deterministic timing.

56 mm

Wheel Diameter

Standard Inventor rubber wheels

112 mm

Axle Track

Distance between drive wheels

200 mm/s

Max Velocity

Recommended cruising speed

Port Mapping & Safety Callout

Connect all peripheral cables to the designated hub ports before powering on:

Hardware Port Assignment

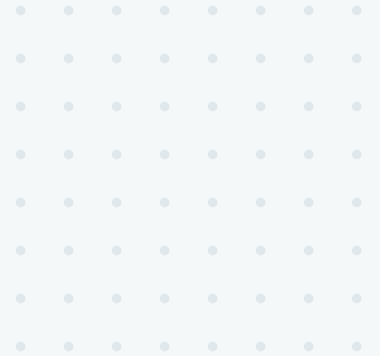
- ◆ **Port.F** : Left Drive Motor
- ◆ **Port.B** : Right Drive Motor
- ◆ **Port.E** : Ground Color Sensor
- ◆ **Port.A** : Ultrasonic Rangefinder
- ◆ **Port.C** : Pitch Scanner Motor
- ◆ **Port.D** : Yaw Scanner Motor

Safety & Motor Limits

- ◆ Never force motors past physical stops while powered on.
- ◆ Set soft software limits:
 - Yaw limits: -115° to -55°
 - Pitch limits: -25° to 25°
- ◆ Keep battery level above 20% to prevent sudden hub reboot.

02

State Machine & Line Tracking



Line Tracking Architecture

A reactive line tracker transitions across distinct operational states:

Follow State

- ◆ Increments (50mm).
- ◆ Checks sensor at 80ms.
- ◆ Line lost → **Backtrack**.

Backtrack State

- ◆ Reverses move history.
- ◆ Inverts straight / turn.
- ◆ Recovers last known line.

Scan State

- ◆ Sweeps yaw motor.
- ◆ Records angles seen.
- ◆ Steers toward peak.

In robotics, open-loop assumptions fail. Always verify sensory feedback before advancing.

— Autonomous Systems Handbook

Pybricks Implementation

```
1  from pybricks.hubs import InventorHub
2  from pybricks.pupdevices import Motor, ColorSensor
3  from pybricks.parameters import Port, Color, Direction
4  from pybricks.robotics import DriveBase
5
6  hub = InventorHub()
7  left_m = Motor(Port.F, Direction.COUNTERCLOCKWISE)
8  right_m = Motor(Port.B)
9  sensor = ColorSensor(Port.E)
10 robot = DriveBase(left_m, right_m, wheel_diameter=56, axle_track=112)
11
12 while True:
13     robot.drive(150, 0) if sensor.color() == Color.GREEN else robot.stop()
```

03

Hands-on Practicum Task

Lab Assignment 01

Task 1: Line Following & Lost Recovery



30 min

★ Hands-on

Port.F, Port.B, Port.E

100 pts

Implement and test the autonomous navigation routine in your robot:

1 Initialize DriveBase & Invert Left Motor

Ensure the left motor uses `Direction.COUNTERCLOCKWISE` so straight motion drives forward.

2 Calibrate Color Sensor Thresholds

Print raw RGB and ambient reflection values on both the white floor and the green track line.

3 Implement Backtracking Deque

Maintain a history of the last 20 actions using `ucollections.deque(maxlen=20)`.

Checkpoint & Demonstration Area

Once your robot completes 2 consecutive laps without manual intervention, signal a TA for scoring.

Evaluation Rubric & Wrap-up

Scoring Criteria

- ◆ Smooth line tracking: **40 pts**
- ◆ Lost state recovery: **30 pts**
- ◆ Clean code & comments: **20 pts**
- ◆ Lab cleanup: **10 pts**

Common Troubleshooting

- ◆ **Robot spins in circles:** Check motor port orientation and direction flag.
- ◆ **Sensor misses line:** Adjust sensor height to 8 — 12mm above ground.
- ◆ **Bluetooth disconnects:** Re-pair using the central connect button.